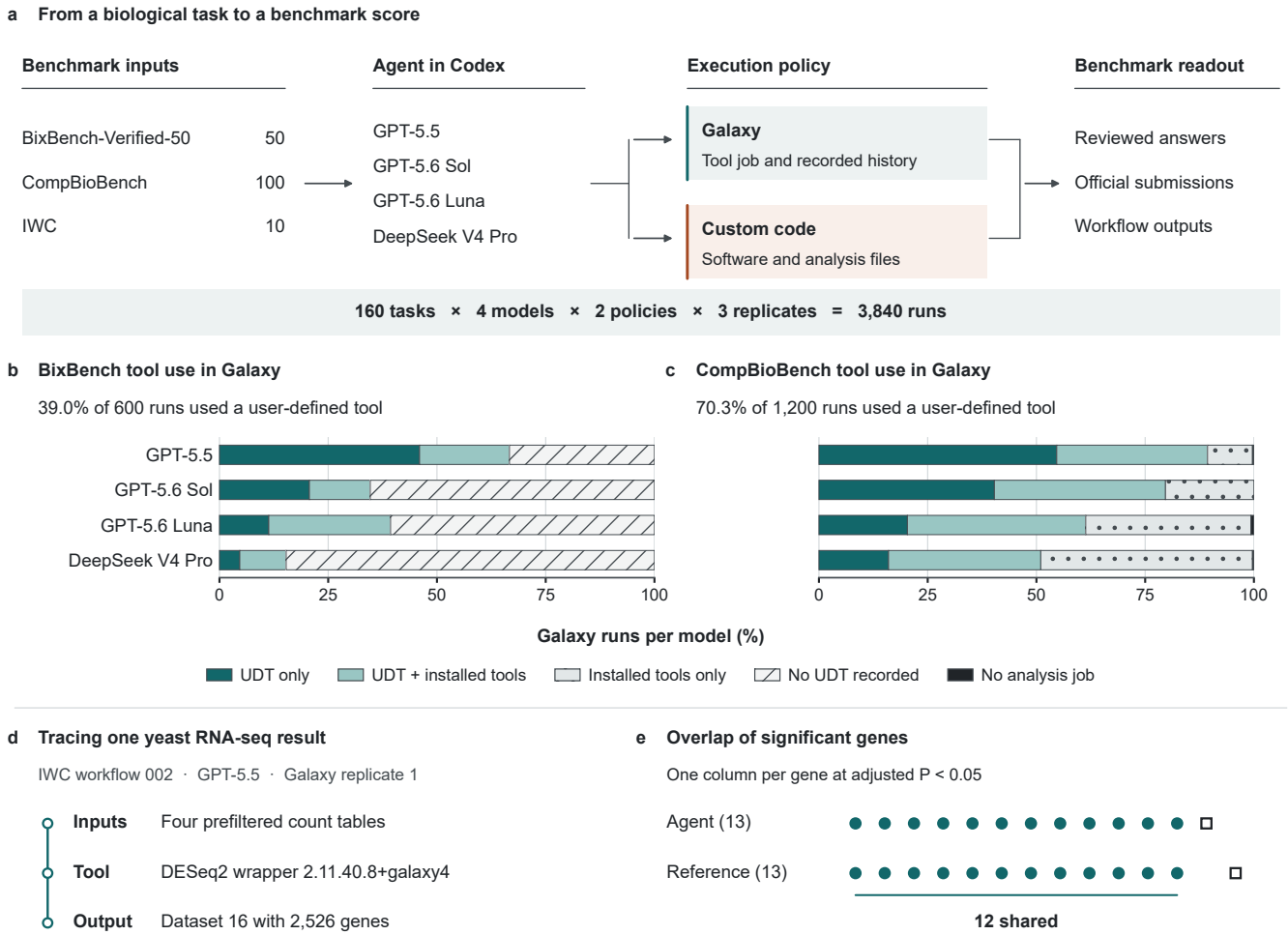
Supplementary Figure 1

Experimental architecture and the provenance of a Galaxy output



a, Four Codex model configurations received the same 160 tasks in Galaxy and custom code, with three replicates of each assignment. Benchmark readouts correspond to the task sets in the order shown. **b,c**, Stacked bars classify 150 BixBench or 300 CompBioBench Galaxy runs per model by observed tool use. A user-defined tool (UDT) packages a custom operation in Galaxy. BixBench records whether a UDT was observed. CompBioBench also distinguishes installed-tool-only histories and histories with no analysis job. **d**, A versioned DESeq2 wrapper processed four prefiltered yeast count tables in a recorded Galaxy job. **e**, Each column represents a gene significant at adjusted $P < 0.05$ in the agent or raw curated output. Filled circles mark shared genes and open squares mark genes unique to one list. The official IWC evaluator uses a prefiltered gene universe.

Supplementary Figure 2

Selection of reruns and effects on reported scores

a **Why 77 runs were selected for replacement**

Reported reason	BixBench	CompBioBench	IWC
Access to an earlier run	11	49	0
Direct copying	3	0	0
Exposure unconfirmed	5	0	0
Scientific follow-up	0	6	0
Reference output access	0	0	3
Replacement runs	**19**	**55**	**3**

b **CompBioBench · change in official score**

Percentage points per 100-answer submission

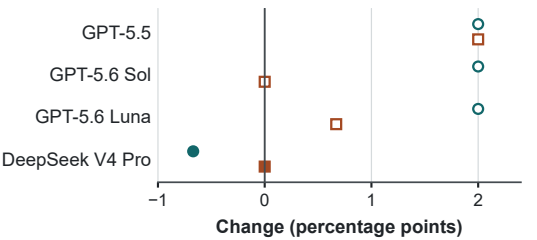
	Galaxy			Custom code		
	R1	R2	R3	R1	R2	R3
GPT-5.5	–	–	0	–	–	–
GPT-5.6 Sol	–	–	–	0	0	–
GPT-5.6 Luna	−1	−1	0	0	−1	−1
DeepSeek V4 Pro	+1	−1	+2	+1	−1	+2

R, replicate · dash, no selected replacement

40% of 55 replacement answers changed.

c **BixBench · change in acceptance**

150 runs per model and policy



Circle, Galaxy · square, custom code

Filled, includes reruns · open, scoring rule only

d **IWC · DeepSeek V4 Pro mean output agreement**

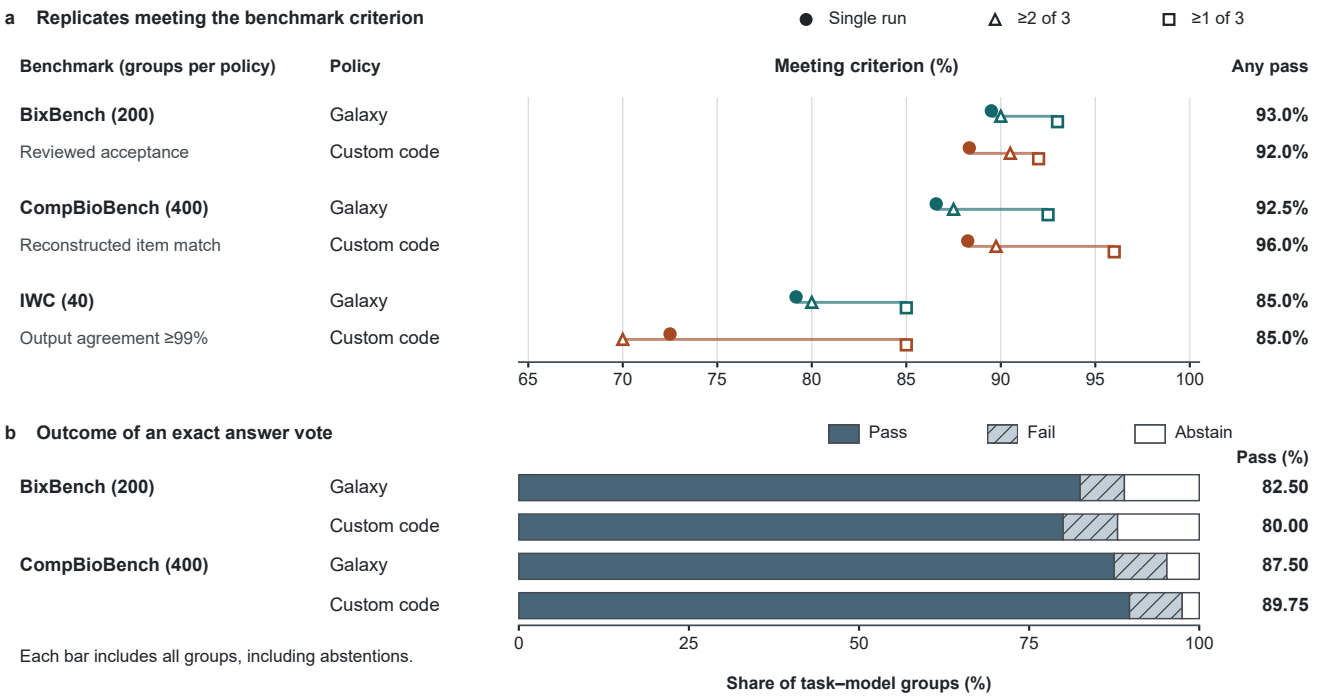
Galaxy 99.94% → 99.88% (1 run replaced out of 30)

Custom code 96.28% → 95.92% (2 runs replaced out of 30)

a, Reasons for selecting individual runs from the previous release, as classified from the 28 October report. Zero denotes no selected run for that reason. **b**, Each shaded cell gives the change in official CompBioBench score in percentage points for a 100-answer submission. Dashes identify unaffected submissions. Forty percent of the 55 replacement answer strings changed. The pooled official means were unchanged in both policies. **c**, BixBench changes include regrading of a PhyKIT question with two accepted version-dependent answers. Open markers identify model–policy groups with zero replacements. Filled markers identify groups with replacements and regrading. Each group comprises 150 runs. **d**, IWC DeepSeek V4 Pro means include three replacements on the peptide-verification task across the two policies. The means summarize 30 workflow runs per policy.

Supplementary Figure 3

Repeated attempts, answer consensus and independent validation



a, Points compare individual runs with groups in which at least two or at least one of three attempts met the criterion. Each group contains one task and one model under one policy. BixBench uses reviewed acceptance, CompBioBench uses reconstructed item matches and IWC uses workflow-output agreement ≥99%. The two group criteria require known grades. **b**, A vote selects a nonempty answer string returned at least twice after trimming whitespace and ignoring case. Each percentage uses all task–model groups under that policy, including abstentions. The selected answer is graded afterwards. IWC outputs have no comparable answer strings. **c**, For one promoter question, the GRCh38 sequence at the strand-aware GENCODE V49 interval independently reproduces the submitted GC fraction. Coordinates are one-based and inclusive.

Supplementary Figure 4

Individual run agreement with each reference workflow

a **Output agreement in each workflow, model and replicate**



b **Workflow summaries**

IWC workflow	Galaxy 5.5 1 2 3	Galaxy Sol 1 2 3	Galaxy Luna 1 2 3	Galaxy DeepSeek 1 2 3	Custom code 5.5 1 2 3	Custom code Sol 1 2 3	Custom code Luna 1 2 3	Custom code DeepSeek 1 2 3	Mean % (runs <99%) Galaxy	Mean % (runs <99%) Custom code
001 Read QC and trimming									**99.89 (0)**	**99.97 (0)**
002 RNA-seq differential expression									**99.95 (0)**	**100.00 (0)**
003 Host-read filtering									**100.00 (0)**	**99.96 (0)**
005 Amplicon denoising					× ×				**95.86 (9)**	**79.67 (9)**
006 ATAC-seq accessibility									**99.78 (0)**	**99.03 (1)**
007 Mitogenome assembly					×				**98.97 (5)**	**90.70 (10)**
008 AMR gene detection									**100.00 (0)**	**100.00 (0)**
009 Peptide verification									**95.64 (10)**	**93.04 (10)**
010 Pseudobulk differential expression								×	**99.79 (1)**	**91.02 (3)**
011 BioProject data retrieval									**100.00 (0)**	**100.00 (0)**

Columns 1–3 are replicates. Models are GPT-5.5, GPT-5.6 Sol, GPT-5.6 Luna and DeepSeek V4 Pro.

Each workflow mean uses 12 runs per policy. A cross marks zero agreement.

a, Each cell denotes one continuous agreement score for an IWC workflow. Within each policy, every model contributed three replicates. The key distinguishes exact agreement, small deviations, larger deviations and zero agreement. Scores retain full source precision for binning. Crosses identify zeroes. **b**, Each entry gives the mean over the 12 runs for that workflow and policy, expressed as a percentage and rounded to two decimals. Parentheses count individual scores strictly below 99% agreement. The diagnostic threshold distinguishes near-matches and is separate from the continuous endpoint in Fig. 1. Different workflows use different output evaluators.

Supplementary Figure 5

Where recorded Galaxy requests failed

a Coverage and stage of historical Galaxy request failures

Benchmark	Traced runs	Before a job ran	During execution	Other	All failures
BixBench	600	895	369	78	1,342
CompBioBench	1,188	2,052	2,650	678	5,380
IWC	108	150	162	64	376

b Failed requests by class and benchmark

Counts are printed in each cell. Shading shows the share of failures within a benchmark.

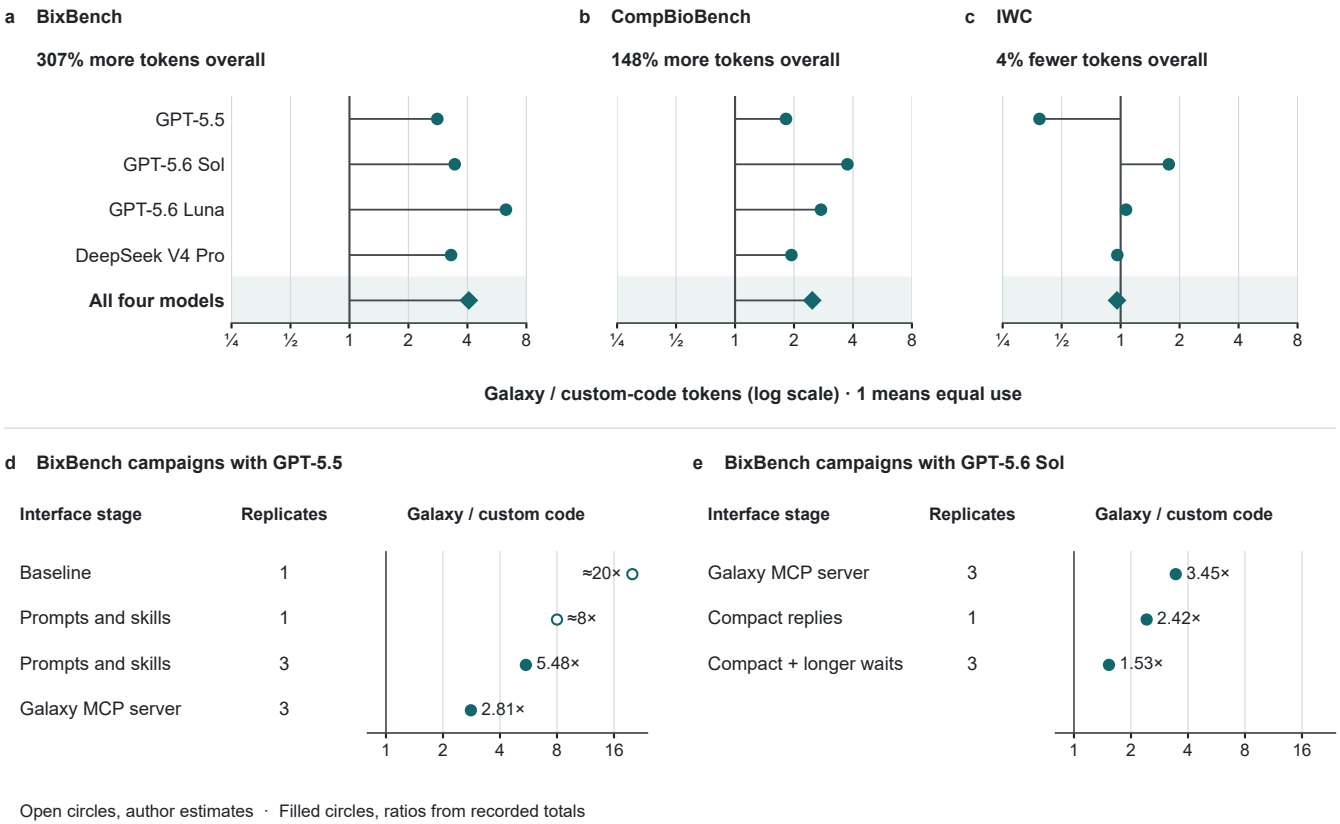
0% | >0–<5% | 5–<10% | 10–<20% | ≥20%

Stage	Failure class	BixBench	CompBioBench	IWC
Before job	Tool schema needs history context	483	590	48
	Tool identifier missing or guessed	49	316	19
	Nested parameter structure	79	123	14
	Parameter value or datatype	150	566	49
	Invalid or foreign identifier	98	132	12
	User-defined tool schema	9	117	0
	Upload or datatype registry	3	91	0
	Server, transport or rate limit	24	117	8
During job	Missing container dependency	12	68	0
	Job error without diagnostic	24	1,242	73
	Tool runtime error	317	1,071	70
	Input format, compression or index	16	176	17
	Memory or resource limit	0	93	2
Other	Additional exceptions	75	458	35
	Unclassified	3	220	29

a, The historical audit covers 600 BixBench, 1,188 CompBioBench and 108 IWC Galaxy runs. Counts refer to failed requests during those traced runs. A run may contribute several failures and still return a correct answer. **b**, Each cell gives the number of requests in one failure class. The shade uses the total failed requests for that benchmark as its denominator. Zeros include classes with no recorded event in that benchmark. A1–A8 in the source precede job execution, B1–B5 concern running jobs, and the other two classes cover exceptions.

Supplementary Figure 6

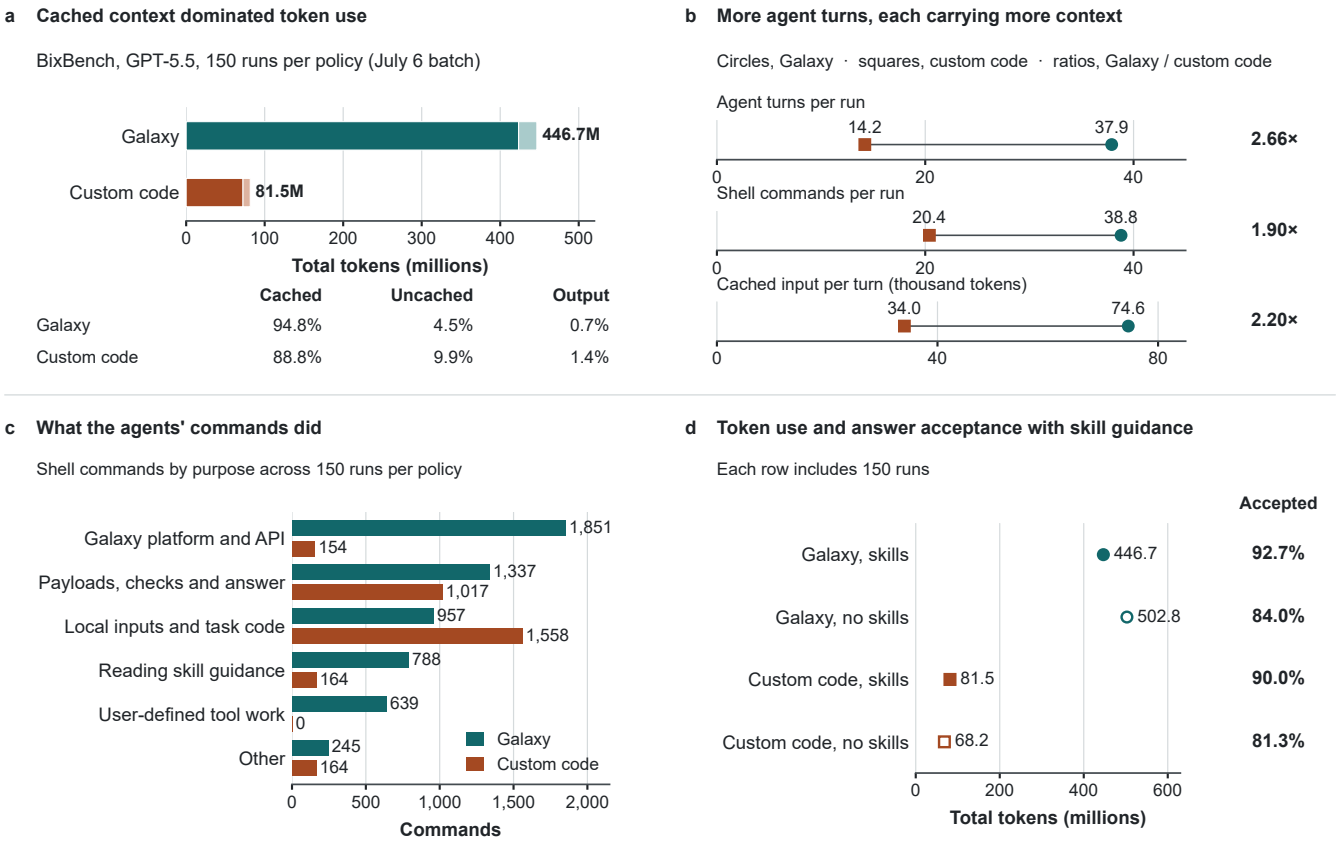
Token use across benchmarks and successive interface versions



a–c, The token ratio divides recorded Galaxy input plus output tokens by the corresponding custom-code total, including cached input once. Points show individual models and diamonds pool token sums. The shared logarithmic scale marks equal token use at one. The headings express the pooled change as a percentage of custom-code token use. One CompBioBench Galaxy run lacked token usage. All other October runs have recorded usage. **d,e**, Rows follow BixBench campaigns in time order within each model. Open circles denote approximate author estimates and filled circles use reported totals. Replicates contain 50 tasks each. The last two GPT-5.6 Sol stages reuse the preceding custom-code baseline. Stages differ in interface behavior, model or CLI version and replicate coverage. They are descriptive comparisons.

Supplementary Figure 7

Where Galaxy runs spent tokens before the interface changes



a, Provider-reported token classes for the July 6 BixBench batch, run with GPT-5.5 in Codex after the first prompt and skill revisions and before the Galaxy MCP server changes. Each policy has 150 runs across 50 tasks and three replicates. Dark bar segments represent cached input and light segments combine uncached input and output. The percentages below the bars separate all three classes. **b**, Agent turns are completed agent messages in the event stream, an observable proxy for model calls within a run. Cached input per turn divides cached input tokens by turns. Ratios are those published with unrounded counts. **c**, Each shell command was assigned one primary purpose by the published rule set. Galaxy platform and API commands include history inspection, tool discovery and schemas, job submission, polling and output retrieval. Command counts are not token counts. **d**, The same batch also ran each policy without skill guidance, including the guidance that told agents to save bulky Galaxy responses to files. Labels give total tokens in millions and the percentage of the 150 answers accepted. These are descriptive comparisons within one batch, model and interface version.